

Supplement to “Limits of Honest Adaptation in a Binary Instrumental-Variable Model”

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This supplement contains the complete lower- and upper-bound proofs, two additional analytic exhibits, and the algebra and measurability details supporting the main article. Equation numbers of the form (4.x), (5.x), (6.x), and (A.x) retain their identifiers from the full v8 manuscript so that the provenance of every calculation remains explicit.

S.1 Full lower-bound proof

This section proves the lower bounds in Theorem 3.1 and Proposition 3.2. Fix $n \geq 2$, $0 < \lambda_n \leq 1/32$, $\alpha \in (0, 1/2)$, and the known $\beta \in (0, 1/4)$. Write $c = c_n = 6\lambda_n$ and retain the reference law P^* of (3.1). We first construct valid IV alternatives with a common target and then bound the distance between their sample mixture and $(P^*)^{\otimes n}$. The calculation uses a second-moment mixture argument and a two-target coverage transfer; neither is a new testing principle. The single-null specialization of Robins et al. (2009), Theorem 2.1, is a precedent for the mixture method; the full-observation calculation below is given directly.

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S.1.1 An IV-compatible perturbation

For binary values z, d, y , write $\sigma_z = 2z - 1$, $\sigma_d = 2d - 1$, and $\sigma_y = 2y - 1$. Given a measurable function h on $[0, 1]$, keep $X \sim U[0, 1]$ and define

$$p_h(z, d, y | x) = \frac{1 + c\sigma_z\sigma_d + h(x)(\sigma_z + \sigma_y)}{8}. \quad (4.1)$$

Here p_h is the joint conditional mass function of (Z, D, Y) given X . At $h = 0$ it is exactly (3.1).

Lemma 4.1 (IV-compatible family). Suppose $\|h\|_\infty \leq 1/4$ and $h(x)^2 \leq c$ almost everywhere. Then (4.1) is an observed law P_h with the overlap, cell-interiority, and monotone-IV properties in Section 2. Its nuisance means and target moments satisfy

$$\zeta_h = \rho_h = \frac{1+h}{2}, \quad \pi_h = \frac{1}{2}, \quad \Lambda(P_h) = \frac{c}{4}, \quad N(P_h) = -\frac{1}{4} \int_0^1 h(x)^2 dx. \quad (4.2)$$

Proof. The eight numerators in (4.1) sum to eight and are at least $1 - c - 2|h| \geq 5/16$. Summing over d, y gives $P_h(Z = z | X = x) = (1 + \sigma_z h)/2$. Thus $\zeta_h \in [3/8, 5/8]$, and the Z -conditional cell probabilities are

$$r_{dy|z,h} = \frac{1 + c\sigma_z\sigma_d + h(\sigma_z + \sigma_y)}{4(1 + \sigma_z h)} \geq \frac{1}{16}. \quad (4.3)$$

The four monotonicity differences, at each x , are

$$\begin{aligned} r_{1y|1,h} - r_{1y|0,h} &= \frac{c - \sigma_y h^2}{2(1 - h^2)}, \\ r_{0y|0,h} - r_{0y|1,h} &= \frac{c + \sigma_y h^2}{2(1 - h^2)}. \end{aligned} \quad (4.4)$$

They are nonnegative exactly when $h^2 \leq c$. Section 2.1 therefore supplies a latent representation with conditional instrument independence, exclusion, and treatment monotonicity. For example, its complier probability and conditional complier effect are $p_{CP} = c/(1 - h^2)$ and $\theta_{CP} = -h^2/c$. The complier probability is not held constant when h changes. A positive first stage alone would not suffice for (4.4).

Summing the cells also gives the three means in (4.2), together with $E_h[ZD | X] = (1 + c + h)/4$ and $E_h[ZY | X] = (1 + 2h)/4$. Subtracting the corresponding products of means proves (4.2). Consequently $\tau(P_h) = -\int h^2/c \in [-1, 0]$. This lemma imposes no smoothness on latent outcome functions; the required regularity of the observed nuisance means is verified next. $\square$

S.1.2 Smooth alternatives with a common target

Let $\varphi(u) = 16u^2(1-u)^2(2u-1)$ on $[0, 1]$, extended by zero elsewhere. The extension is continuously differentiable, with zero value and derivative at both endpoints, and

$$\int_0^1 \varphi(u) du = 0, \quad \mu_2 := \int_0^1 \varphi(u)^2 du = \frac{128}{3465}, \quad \|\varphi\|_\infty \leq 1, \quad \text{Lip}(\varphi) \leq 2. \quad (4.5)$$

The zero integral follows from antisymmetry about $1/2$, and the second moment follows by integrating $256u^4(1-u)^4(2u-1)^2$. For the derivative bound, put $t = u(1-u) \in [0, 1/4]$; then $\varphi'(u) = 32t(5t-1) \in [-8/5, 2]$. No infinitely differentiable bump or numerical grid bound is required.

Divide each summary cell into J equal blocks, so that $m = 2J$ blocks partition $[0, 1]$. For a sign vector $v \in \{-1, 1\}^m$, define

$$h_v(x) = \delta v_j \varphi(mx - j), \quad x \in [j/m, (j+1)/m], \quad j = 0, \dots, m-1. \quad (4.6)$$

The definitions agree at block boundaries because the bump vanishes there. Set

$$\gamma_\alpha = \frac{1-2\alpha}{4}, \quad a_\beta = \frac{2}{4\beta+1}, \quad J = \lceil n^{a_\beta} \rceil, \quad \omega = \delta^2 = \gamma_\alpha \min\{c, J^{-2\beta}\}. \quad (4.7)$$

In particular, $0 < \omega \leq \gamma_\alpha c < 3/64 < 1/16$ and $\omega \leq c$, so Lemma 4.1 applies to every sign vector. These are sufficient amplitude restrictions, not sharp necessary restrictions on δ . The

construction specifies probability laws and remains valid when $J > n$; it does not require a sample Gram matrix to be invertible.

To check the exact Hölder convention in (2.9), let $h_{v,s}(u) = h_v((s+u)/2)$ for $s \in \{0, 1\}$. The cellwise Lipschitz constant is at most $2\delta J$, including across block boundaries. Combining this bound with $\|h_v\|_\infty \leq \delta$ gives, for any $u, u' \in [0, 1]$,

$$|h_{v,s}(u) - h_{v,s}(u')| \leq 2\delta \min\{1, J|u - u'|\} \leq 2\delta J^\beta |u - u'|^\beta. \quad (4.8)$$

Hence the Hölder seminorms of ζ_h and ρ_h are at most $\delta J^\beta \leq \sqrt{\gamma_\alpha} < 1/2$, and that of π_h is zero. Their sup norms are also below two. This verifies membership in the stated class for every sign vector, every block boundary, and every real $\beta \in (0, 1/4)$, not merely a finite set of smoothness indices.

Write $P_v = P_{h_v}$. Every block has zero integral, while $\int_0^1 h_v^2 = \mu_2 \omega$. Thus all components belong to $\mathcal{P}_{1,n}(\beta) \setminus \mathcal{P}_{0,n}$ and satisfy

$$\Lambda(P_v) = \Lambda(P^*) = \frac{c}{4} = \frac{3}{2}\lambda_n, \quad \tau(P_v) = -\Delta, \quad \Delta = \frac{\mu_2 \omega}{c} = \mu_2 \gamma_\alpha \min\{1, J^{-2\beta}/c\}. \quad (4.9)$$

The exclusion from $\mathcal{P}_{0,n}$ follows because $\delta > 0$ and h_v is nonconstant within each summary cell. Its mean within each cell is nevertheless zero. The target separation is positive and independent of v , and the identification stratum is the same as at the null. In particular, weak identification constrains the admissible perturbation through $\omega \leq c$; the separation is not obtained by dividing an unrestricted signal by a vanishing denominator.

S.1.3 Distance of the sample mixture

Choose v uniformly once and retain it for all n observations. Define the sample law

$$Q_n = 2^{-m} \sum_{v \in \{-1,1\}^m} P_v^{\otimes n}. \quad (4.10)$$

Although the one-observation mixture is $2^{-m} \sum_v P_v = P^*$, its n -fold product is not the sample mixture in (4.10). We do not assume Q_n is an iid member of the causal model and do not assign it a causal parameter $\tau(Q_n)$.

Use $\text{TV}(Q, P) = \sup_A |Q(A) - P(A)| = \frac{1}{2} \int |dQ - dP|$ and $\chi^2(Q, P) = \int (dQ/dP - 1)^2 dP$ when $Q \ll P$.

Lemma 4.2 (Shared-sign mixture bound). For the choices in (4.7),

$$1 + \chi^2(Q_n, (P^*)^{\otimes n}) \leq \exp(x_n), \quad x_n = \frac{n^2 \xi_c^2 \omega^2}{4J}, \quad \xi_c = \frac{2\mu_2}{1-c^2} < 1, \quad (4.11)$$

and $\text{TV}(Q_n, (P^*)^{\otimes n}) \leq \gamma_\alpha$.

Proof. Since $c \leq 3/16$, $\xi_c \leq 65536/855855 < 1$. With $\sigma_Z = 2Z - 1$ and the analogous random signs for D, Y , the one-observation likelihood ratio is

$$\ell_v(W) = \frac{dP_v}{dP^*}(W) = 1 + h_v(X) \frac{\sigma_Z + \sigma_Y}{1 + c\sigma_Z\sigma_D}. \quad (4.12)$$

The random fraction has conditional mean zero under P^* . Its conditional second moment is $\sum_{\sigma_z, \sigma_d, \sigma_y} (\sigma_z + \sigma_y)^2 / \{8(1 + c\sigma_z\sigma_d)\} = 2/(1 - c^2)$, where the sum runs over all eight sign triples. Therefore, for two sign vectors v, v' ,

$$E_{P^*}[\ell_v \ell_{v'}] = 1 + \frac{\xi_c \omega}{m} \sum_{j=0}^{m-1} v_j v'_j. \quad (4.13)$$

Independence of observations within each component and expansion of the finite mixture give

$$\begin{aligned} 1 + \chi^2(Q_n, (P^*)^{\otimes n}) &= E_{v, v'} \left(1 + \frac{\xi_c \omega}{m} \sum_j v_j v'_j \right)^n \\ &\leq \left\{ \cosh \left(\frac{n \xi_c \omega}{m} \right) \right\}^m \leq \exp \left(\frac{n^2 \xi_c^2 \omega^2}{2m} \right). \end{aligned} \quad (4.14)$$

Indeed, under independent uniform vectors v, v' , their coordinatewise products are independent fair signs. Every parenthesis is positive because $\xi_c \omega < 1$. The first inequality uses $\log(1+t) \leq t$, and the second uses $\log \cosh(t) \leq t^2/2$, obtained by integrating $\tanh(t) \leq t$ for $t \geq 0$ and using symmetry. Since $m = 2J$, (4.14) proves (4.11).

Now $\omega \leq \gamma_\alpha J^{-2\beta}$ and $J \geq n^{2/(4\beta+1)}$ imply $x_n \leq \gamma_\alpha^2/4 < 1/64$. Cauchy–Schwarz and the elementary bound $e^x - 1 \leq x/(1-x)$ for $0 \leq x < 1$ yield

$$\text{TV}(Q_n, (P^*)^{\otimes n})^2 \leq \frac{e^{x_n} - 1}{4} \leq \frac{x_n}{4(1-x_n)} \leq \frac{4\gamma_\alpha^2}{63} < \gamma_\alpha^2. \quad (4.15)$$

All distance bounds concern the full observations, including D and X . $\square$

The family also admits a useful reduction that clarifies the role of the IV embedding. Put $g = h/(1-c)$ and consider two signs with conditional law $R_g(A=a, E=e \mid X=x) = \{1 + g(x)(a+e)\}/4$. This is positive since $|g| \leq 4/13 < 1/2$. A transformation K_c , independent of h , keeps the two input signs and appends an independent fair treatment sign with probability $1-c$. With probability c , it discards the input signs, draws fresh independent fair instrument and outcome signs, and sets the treatment sign equal to the instrument sign. The output has mass

$$(1-c) \frac{1 + g(x)(\sigma_z + \sigma_y)}{8} + c \frac{1 + \sigma_z \sigma_d}{8} = p_h(z, d, y \mid x). \quad (4.16)$$

Applying this same transformation independently to observations cannot increase total variation, including for the shared-sign sample mixtures. It is not an equivalence of experiments, and the transformation depends on the fixed subexperiment scale c , not on an unknown perturbation. We use the direct likelihood proof above; the alternative source-distance calculation is given in Supplement Section S.7. The reduction supplies no new estimation information or new testing principle.

S.1.4 Transfer of coverage and the theorem's constant

The two-target coverage-transfer principle is standard; see Cai and Guo (2015, version 2), Lemma 1. We give the argument here for the confidence-set class of Section 2.4, including auxiliary randomization.

Let $C_n \in \mathfrak{C}_{\alpha,n}(\mathcal{P}_{1,n}(\beta))$, with any auxiliary randomization $U \sim \nu$ from Section 2.4. Since every component has target $-\Delta$, averaging its coverage gives

$$(Q_n \otimes \nu)\{-\Delta \in C_n\} \geq 1 - \alpha. \quad (4.17)$$

This applies honesty to the individual P_ν , not to Q_n as a new causal law. Taking a product with the common independent law ν cannot increase total variation. Lemma 4.2 and honesty at P^* therefore imply

$$\mathbb{P}_{P^*,n}\{-\Delta \in C_n\} \geq 1 - \alpha - \gamma_\alpha, \quad \mathbb{P}_{P^*,n}\{0 \in C_n\} \geq 1 - \alpha. \quad (4.18)$$

The probability of joint inclusion is at least $1 - 2\alpha - \gamma_\alpha = 3(1 - 2\alpha)/4$. A set containing both points has diameter at least Δ , regardless of connectedness. Hence

$$\begin{aligned} \mathbb{E}_{P^*,n}[\text{diam}(C_n)] &\geq \Delta(1 - 2\alpha - \gamma_\alpha) \\ &= \frac{3\mu_2(1 - 2\alpha)^2}{16} \min\{1, J^{-2\beta}/(6\lambda_n)\}. \end{aligned} \quad (4.19)$$

To account for integer resolution, $n^{a\beta} \leq J \leq 2n^{a\beta}$ and $\beta < 1/4$ give $J^{-2\beta} \geq 2^{-2\beta}n^{-r\beta} \geq n^{-r\beta}/2$.

In particular,

$$\min\{1, J^{-2\beta}/(6\lambda_n)\} \geq \frac{1}{12} \min\{1, n^{-r\beta}/\lambda_n\}. \quad (4.20)$$

Substituting into (4.19) proves

$$\mathbb{E}_{P^*,n}[\text{diam}(C_n)] \geq \frac{\mu_2(1 - 2\alpha)^2}{64} q_{\beta,n} = \underline{k}_\alpha q_{\beta,n}, \quad (4.21)$$

with precisely the constant in (3.4). This proves Theorem 3.1(i) for every allowed n, λ_n, β and includes empty and disconnected confidence sets. No asymptotic coverage remainder, restriction to compressed data, or fixed-resolution illustration is used. The large-model upper bound is a separate argument in Supplement Section S.2.

S.1.5 The oracle lower bound at the same reference law

For $|\theta| \leq 1/2$, define a latent IV family P_θ^{or} as follows, conditionally on each x : take $\zeta = 1/2$, $p_{\text{CP}} = c$, and $p_{\text{AT}} = p_{\text{NT}} = (1 - c)/2$. Set

$$q_{\text{AT},1} = q_{\text{NT},0} = \frac{1}{2}, \quad q_{\text{CP},0} = \frac{1 - \theta}{2}, \quad q_{\text{CP},1} = \frac{1 + \theta}{2}. \quad (4.22)$$

Choose the unobserved outcome margins to be Bernoulli($1/2$) and a product coupling within each type, and generate Z independently of the latent variables conditional on X as in Section 2.1. This constructs a submodel, without adding a potential-outcome independence assumption to the full model. Its three nuisance means are $1/2$, every observed Z -conditional cell is at least $13/64$, and $\Lambda = c/4$, $\tau = \theta$. Thus $P_\theta^{\text{or}} \in \mathcal{P}_{0,n}$, and $P_0^{\text{or}} = P^*$.

Relative to P^* , four joint cells change by $\pm c\theta/4$, each with null probability $(1 + c)/8$. The other cells are unchanged. Therefore

$$\chi^2(P_\theta^{\text{or}}, P^*) = \frac{2c^2\theta^2}{1 + c}, \quad 1 + \chi^2((P_\theta^{\text{or}})^{\otimes n}, (P^*)^{\otimes n}) = \left(1 + \frac{2c^2\theta^2}{1 + c}\right)^n. \quad (4.23)$$

Put $\varepsilon_\alpha = (1 - 2\alpha)/2$ and choose

$$b_\alpha = \sqrt{\frac{\log(1 + 4\varepsilon_\alpha^2)}{72}}, \quad \theta_n = \min\{1/2, b_\alpha/(\sqrt{n}\lambda_n)\}. \quad (4.24)$$

Since $c = 6\lambda_n$, the exponent bound $2nc^2\theta_n^2/(1 + c) \leq 72b_\alpha^2$ implies that the sample total variation distance is at most ε_α . For every procedure honest on $\mathcal{P}_{0,n}$, the same two-target argument at 0 and θ_n yields

$$\mathbb{E}_{P^*,n}[\text{diam}(C_n)] \geq \theta_n(1 - 2\alpha - \varepsilon_\alpha) \geq \underline{k}_{0,\alpha} q_{0,n}, \quad \underline{k}_{0,\alpha} := \frac{1 - 2\alpha}{2} \min\{1/2, b_\alpha\} > 0. \quad (4.25)$$

The last inequality uses $\min\{a, b/x\} \geq \min\{a, b\} \min\{1, 1/x\}$ for positive a, b, x . This establishes the lower bound of Proposition 3.2 at exactly the reference law used in Theorem 3.1. The oracle upper bound, which requires no nuisance values, remains the two-cell construction in Supplement Section S.2.

S.2 Full confidence-set construction and upper-bound proof

We construct a confidence set attaining the upper bound in Theorem 3.1 and then obtain the oracle upper bound in Proposition 3.2. The construction uses projection estimates of the two moments in (2.6), a deterministic allowance for their approximation bias, and inversion of the moment equation. Its guarantees hold on the entire model of Section 2, not just on the alternatives used in Supplement Section S.1. The procedure is deterministic given the sample and thus belongs to the class allowing auxiliary randomization. Projection-based covariance U-statistics and covariance-based inversion have a precedent in Li et al. (2011), Sections 1, 3, and 6. We use that strategy in the present IV model and prove the stated bounds directly.

S.2.1 Projection moments and their bias

Initially, let $J \geq 1$ be any integer fixed before observing the data. Partition each summary cell into J equal bins, for a total of $2J$ bins $I_{J,k}$, $k = 0, \dots, 2J - 1$. Use left-closed, right-open bins, except that the last bin includes 1. Endpoints have probability zero. Each bin has known probability $1/(2J)$. Write $\langle f, g \rangle = \int_0^1 f(x)g(x) dx$ and let Π_J be the orthogonal projection onto functions constant on these bins:

$$K_J(x, x') = 2J \sum_{k=0}^{2J-1} \mathbf{1}_{I_{J,k}}(x) \mathbf{1}_{I_{J,k}}(x'), \quad (\Pi_J f)(x) = \int_0^1 K_J(x, x') f(x') dx'. \quad (5.1)$$

The normalized bin indicators have population Gram matrix equal to the identity. This known projection does not involve an empirical Gram inverse, including when many bins contain no observations. Knowing the projection does not reveal the propensity or the other conditional means.

For $V \in \{Y, D\}$, put $\mu_V(x) = E_P[V \mid X = x]$ and $T_V = E_P[ZV] - \langle \zeta, \mu_V \rangle$, so that $T_Y = N$ and $T_D = \Lambda$. Define

$$\widehat{T}_{V,J} = \frac{1}{n} \sum_{i=1}^n Z_i V_i - \frac{1}{n(n-1)} \sum_{i \neq j} K_J(X_i, X_j) Z_i V_j, \quad \widehat{N}_J = \widehat{T}_{Y,J}, \quad \widehat{\Lambda}_J = \widehat{T}_{D,J}. \quad (5.2)$$

Only off-diagonal pairs enter the second term. Both moment estimates use the same n observations; no independence between them is assumed.

Lemma 5.1 (Projection bias). For every $P \in \mathcal{P}_{1,n}(\beta)$ and fixed integer $J \geq 1$,

$$E_P \widehat{T}_{V,J} = T_V + B_{V,J}, \quad B_{V,J} = \langle (I - \Pi_J) \zeta, (I - \Pi_J) \mu_V \rangle, \quad |B_{V,J}| \leq b_{\beta,J} := \min\{1/4, 4J^{-2\beta}\}. \quad (5.3)$$

Here and below, E_P in moment calculations denotes expectation under the indicated iid sample, or its one- or two-observation marginal as appropriate.

Proof. Independence of two distinct observations and orthogonality of the projection give

$$E_P[K_J(X_i, X_j) Z_i V_j] = \langle \Pi_J \zeta, \Pi_J \mu_V \rangle, \quad \langle \zeta, \mu_V \rangle = \langle \Pi_J \zeta, \Pi_J \mu_V \rangle + \langle (I - \Pi_J) \zeta, (I - \Pi_J) \mu_V \rangle. \quad (5.4)$$

Subtracting the target from the mean of (5.2) proves the bias identity, including its sign. Within a bin, the width in the cell coordinate of (2.9) is $1/J$. Consequently, for either relevant conditional mean f ,

$$\|f - \Pi_J f\|_2 \leq 2J^{-\beta}, \quad \|f - \Pi_J f\|_2^2 = E_P[\text{Var}_P(f(X) \mid \text{bin}_J(X))] \leq \frac{1}{4}. \quad (5.5)$$

The first inequality uses the cellwise Hölder bound; the second uses $0 \leq f \leq 1$. The two Cauchy–Schwarz bounds yield (5.3). $\square$

The allowance $b_{\beta,J}$ follows from the specified function class. It is not an estimate or a data-based certificate of the actual bias. Although the bias vanishes at simple laws, the full-model procedure below does not remove this allowance on the basis of an apparent fit to the simple model.

S.2.2 Variance without a restriction on the number of bins

Lemma 5.2 (Full projection variance). For every law in the large model, every $n \geq 2$, and every fixed integer $J \geq 1$,

$$\text{Var}_P(\widehat{T}_{V,J}) \leq v_{n,J} := \frac{1}{n} + \frac{2(1+2J)}{n(n-1)}. \quad (5.6)$$

Proof. The estimator is an order-two U-statistic with symmetric kernel $H_{V,J}$. For an independent observation W' , its one-observation conditional mean is

$$\begin{aligned} H_{V,J}(w, w') &= \frac{zv + z'v'}{2} - \frac{K_J(x, x')}{2}(zv' + z'v), \\ H_{V,J}^{(1)}(w) &= E_P[H_{V,J}(w, W')] = \frac{zv + E_P[ZV] - z(\Pi_J \mu_V)(x) - v(\Pi_J \zeta)(x)}{2}. \end{aligned} \quad (5.7)$$

Let $\vartheta_{V,J} = E_P[H_{V,J}(W, W')]$, $G_1(w) = H_{V,J}^{(1)}(w) - \vartheta_{V,J}$, and $G_2(w, w') = H_{V,J}(w, w') - H_{V,J}^{(1)}(w) - H_{V,J}^{(1)}(w') + \vartheta_{V,J}$. Thus $E_P[G_2(W, W') \mid W] = 0$. The orthogonal decomposition gives

$$\begin{aligned} \widehat{T}_{V,J} - \vartheta_{V,J} &= \frac{2}{n} \sum_i G_1(W_i) + \binom{n}{2}^{-1} \sum_{i < j} G_2(W_i, W_j), \\ \text{Var}_P(\widehat{T}_{V,J}) &= \frac{4}{n} E_P G_1^2 + \frac{2}{n(n-1)} E_P G_2^2. \end{aligned} \quad (5.8)$$

Distinct degenerate pairs are uncorrelated even when they share one observation, by condition-

ing on that observation. They are also orthogonal to the first-order sum. This explains the second variance term rather than treating the pairs as independent observations.

Since $z, v \in \{0, 1\}$ and both projected means are in $[0, 1]$, the quantity $zv - z\Pi_J\mu_V - v\Pi_J\zeta$ is in $[-1, 1]$. Hence $H_{V,J}^{(1)}$ lies in an interval of length one. Also write $H_{V,J} = A - B$, where the two terms in (5.7) are nonnegative, $A \leq 1$, and $B \leq K_J$. Then $(A - B)^2 \leq A^2 + B^2$. Using the bin probabilities and the orthogonal decomposition of the kernel,

$$E_P G_1^2 \leq \frac{1}{4}, \quad E_P K_J(X, X')^2 = 2J, \quad E_P G_2^2 \leq \text{Var}_P(H_{V,J}) \leq E_P H_{V,J}^2 \leq 1 + 2J. \quad (5.9)$$

Substitution into (5.8) proves (5.6). $\square$

This argument uses only iid sampling, bounded binary variables, and the known covariate distribution. It does not impose smoothness on conditional variances or the joint outcome-treatment law. In particular, it remains valid for $J > n$; the contribution of order J/n^2 is retained. No central limit theorem or first-order score approximation is used.

S.2.3 Bias-aware inversion and finite-sample coverage

Bias-aware inversion for weakly identified ratios is an established approach; see Noack and Rothe (2023, version 4), Section 3, in fuzzy regression discontinuity. The finite-sample calculation below uses the different model and moment bounds specified here.

For the chosen fixed J , abbreviate $b = b_{\beta,J}$ and $v = v_{n,J}$. Choose the deterministic upward dyadic rounding

$$e_{n,J,\alpha} = 2^{\lceil \log_2 \sqrt{2v_{n,J}/\alpha} \rceil}, \quad \sqrt{2v/\alpha} \leq e_{n,J,\alpha} < 2\sqrt{2v/\alpha}, \quad R_{\beta,n,J} = b + e_{n,J,\alpha}. \quad (5.10)$$

Integer powers may be negative. The relative rounding bound, including equality at exact powers of two, suffices for the rate calculation. This definition specifies a mathematical procedure;

it does not claim exact finite-precision evaluation of arbitrary real β . Write $e = e_{n,J,\alpha}$ and $R = R_{\beta,n,J}$ within this subsection and the next. Define

$$C_{n,J}^\beta = \{t \in [-1, 1] : |\hat{N}_J - t\hat{\Lambda}_J| \leq R(1 + |t|)\}. \quad (5.11)$$

Proposition 5.3 (Finite-sample coverage). For every fixed integer $J \geq 1$, the set in (5.11) is admissible and

$$\inf_{P \in \mathcal{P}_{1,n}(\beta)} \mathbb{P}_{P,n}\{\tau(P) \in C_{n,J}^\beta\} \geq 1 - \alpha. \quad (5.12)$$

Proof. Lemma 5.2 and a union bound give

$$\mathbb{P}_{P,n}\left\{\max_{V \in \{Y,D\}} |\hat{T}_{V,J} - E_P \hat{T}_{V,J}| > e\right\} \leq \frac{2v}{e^2} \leq \alpha. \quad (5.13)$$

On the complementary event, Lemma 5.1 bounds each moment's error relative to its true value by R . Since $N = \tau\Lambda$ and $\tau \in [-1, 1]$, the triangle inequality gives $|\hat{N}_J - \tau\hat{\Lambda}_J| \leq R(1 + |\tau|)$. The constants do not depend on P .

For admissibility, (5.11) is closed in the compact parameter space. More explicitly, on $[0, 1]$ take $\eta = 1$, and on $[-1, 0]$ take $\eta = -1$, so that $|t| = \eta t$. The acceptance condition becomes

$$(-\hat{\Lambda}_J - R\eta)t \leq R - \hat{N}_J, \quad (\hat{\Lambda}_J - R\eta)t \leq R + \hat{N}_J. \quad (5.14)$$

Each half contributes a closed interval, a singleton, or the empty set. Its endpoints and feasibility are measurable functions of the estimates, obtained by intersecting half-lines and handling zero coefficients separately. Thus the union is a measurable closed set with measurable diameter and coverage events. At a zero coefficient, the associated inequality is either vacuous or infeasible according to its right-hand side; no perturbation of that coefficient is required. $\square$

The set can be empty or disconnected. Its diameter is the span of all accepted points, not the sum of the lengths of its components, and empty outcomes remain in the coverage and risk

calculations. The construction neither clips a small denominator nor divides by its estimate. The argument is uniform over candidate t at this fixed resolution, not a simultaneous guarantee over a data-selected family of resolutions.

S.2.4 Expected diameter and the matching rate

We first prove a fixed-resolution bound:

$$\sup_{P \in \mathcal{P}_{1,n}(\beta)} \mathbb{E}_{P,n}[\text{diam}(C_{n,J}^\beta)] \leq \min \left\{ 2, \frac{8R}{\lambda_n} + \frac{32v}{\lambda_n^2} \right\} \leq 16 \min\{1, R/\lambda_n\}. \quad (5.15)$$

Suppose first that $\lambda_n \geq 4b$. For any law in the model, $\Lambda \geq \lambda_n$ and $E_P \hat{\Lambda}_J \geq \Lambda - b \geq 3\Lambda/4$. Therefore

$$\mathbb{P}_{P,n}\{\hat{\Lambda}_J < \Lambda/2\} \leq \mathbb{P}_{P,n}\{|\hat{\Lambda}_J - E_P \hat{\Lambda}_J| > \Lambda/4\} \leq \frac{16v}{\Lambda^2}. \quad (5.16)$$

On the event $\hat{\Lambda}_J \geq \Lambda/2$, any two accepted points satisfy

$$|t_1 - t_2| \hat{\Lambda}_J \leq |\hat{N}_J - t_1 \hat{\Lambda}_J| + |\hat{N}_J - t_2 \hat{\Lambda}_J| \leq R(2 + |t_1| + |t_2|) \leq 4R. \quad (5.17)$$

The diameter is consequently at most $8R/\Lambda$ on that event. On its complement it is at most two. Taking expectations proves the first inequality in (5.15), also for disconnected or empty sets. If instead $\lambda_n < 4b$, then $8R/\lambda_n > 2$, and the same inequality follows directly from the deterministic diameter bound.

To prove the second inequality, put $x = R/\lambda_n$ and use $v \leq \alpha e^2/2 \leq \alpha R^2/2$. For $0 \leq x \leq 1$, $8x + 16\alpha x^2 \leq 16x$ because $\alpha < 1/2$; for $x \geq 1$, the bound by two suffices. Thus

$$\min\{2, 8R/\lambda_n + 32v/\lambda_n^2\} \leq \min\{2, 8x + 16\alpha x^2\} \leq 16 \min\{1, x\}. \quad (5.18)$$

This expected-diameter argument is separate from coverage. Retaining an uncontrolled event of probability α would leave an additive constant and would not prove contraction. Equation (5.16)

instead controls the bad-denominator contribution at the scale needed for the risk bound.

Now use the same resolution as in Theorem 3.1 and Supplement Section S.1:

$$J_n = \lceil n^{a_\beta} \rceil, \quad a_\beta = \frac{2}{4\beta + 1}, \quad C_n^\beta = C_{n,J_n}^\beta. \quad (5.19)$$

This is fixed by n and the known β before seeing the data. In the stated range, $a_\beta > 1$, so the variance calculation for resolutions exceeding n is essential. Since $n - 1 \geq n/2$, $J_n \leq 2n^{a_\beta}$, $a_\beta - 2 = -2r_\beta$, and $r_\beta < 1/2$,

$$\begin{aligned} v_{n,J_n} &\leq n^{-1} + 4n^{-2} + 8J_n/n^2 \leq 21n^{-2r_\beta}, \\ b_{\beta,J_n} &\leq 4n^{-r_\beta}, \\ R_{\beta,n,J_n} &\leq \left(4 + 2\sqrt{42/\alpha}\right) n^{-r_\beta}. \end{aligned} \quad (5.20)$$

The constant 21 follows by bounding the three variance terms by 1, 4, and 16 times n^{-2r_β} . Combining (5.12), (5.15), and (5.20) gives

$$\sup_{P \in \mathcal{P}_{1,n}(\beta)} \mathbb{E}_{P,n}[\text{diam}(C_n^\beta)] \leq 16 \left(4 + 2\sqrt{42/\alpha}\right) \min\{1, n^{-r_\beta}/\lambda_n\}. \quad (5.21)$$

Here we used $\min\{1, Az\} \leq A \min\{1, z\}$ for $A \geq 1$. This is exactly the constant $\bar{k}_\alpha$ in (3.4), proving Theorem 3.1(ii). Together with Supplement Section S.1 it yields the three matching risks in (3.7). Although λ_n enters the risk bound, neither λ_n nor the unknown $\Lambda(P)$ is needed to calculate this confidence set.

The result is rate attainment by a projection-and-inversion construction, not a new estimation principle or a sharp-constant statement. The displayed upper bound can exceed the trivial bound of two at relevant sample sizes. Its validity does not by itself establish practically short confidence sets or improvement from generated instruments.

S.2.5 The zero-bias oracle upper bound

Suppose now that coverage is required only on $\mathcal{P}_{0,n}$. With $J = 1$, the bins are precisely the two summary cells. The three nuisance means are S -measurable, so $\Pi_1 \zeta = \zeta$ and $\Pi_1 \mu_V = \mu_V$. The bias identity in (5.3) is therefore exactly zero. This conclusion does not require the conditional joint distribution of (Y, D, Z) to be S -measurable.

Use the estimates in (5.2) at $J = 1$ and define

$$\begin{aligned} v_{n,1} &= \frac{1}{n} + \frac{6}{n(n-1)} \leq \frac{7}{n}, \\ R_{0,n} &= e_{n,1,\alpha} < 2\sqrt{14/\alpha} n^{-1/2}, \\ C_n^0 &= \{t \in [-1, 1] : |\hat{N}_1 - t\hat{\Lambda}_1| \leq R_{0,n}(1 + |t|)\}. \end{aligned} \tag{5.22}$$

The subscript 0 refers to the oracle model, not to $\beta = 0$. Here the bias allowance is set to zero because membership in the small model is supplied as oracle information; this is not the full-model allowance $b_{\beta,1}$ and is not selected by a test of simplicity.

The coverage proof applies with $b = 0$. So does the expected-diameter proof: the condition $\lambda_n \geq 4b$ then holds automatically. Consequently,

$$\begin{aligned} \inf_{P \in \mathcal{P}_{0,n}} \mathbb{P}_{P,n}\{\tau(P) \in C_n^0\} &\geq 1 - \alpha, \\ \sup_{P \in \mathcal{P}_{0,n}} \mathbb{E}_{P,n}[\text{diam}(C_n^0)] &\leq 16 \min\{1, R_{0,n}/\lambda_n\} \\ &\leq \bar{k}_{0,\alpha} \min\{1, n^{-1/2}/\lambda_n\}, \quad \bar{k}_{0,\alpha} = 32\sqrt{14/\alpha}. \end{aligned} \tag{5.23}$$

This supplies one admissible α -only upper constant in Proposition 3.2. With the lower constant in (4.25), it completes that proposition at the same reference law P^* . The oracle still estimates unknown nuisance-related moments from the original sample. It receives neither their true values, a known propensity, nor additional observations. The difference between (5.21) and (5.23) is the coverage model under which the bias allowance may be removed, not a change in the causal target or the observed experiment.

S.3 Additional illustrations and finite-sample limitations

S.3.1 A population example of information lost by compression

Consider the valid binary-IV family P_h in (4.1). Fix $\beta = 1/8$, $c = 3/160$, and the identification scale $\lambda_n \equiv 1/320$. Use $J = 2$ bins per summary cell, hence four blocks in total, with signs $(+, -, +, -)$ in (4.6). Compare the null amplitude $\delta = 0$ with the alternative amplitude $\delta = 1/8$. The latter satisfies $\delta \leq 1/4$, $\delta^2 = 1/64 \leq c$, and $\delta^8 J = 1/8388608 \leq 1$. The last inequality implies $\delta J^\beta \leq 1$, which is sufficient for the cellwise Hölder condition. Lemma 4.1 and the bump bounds therefore place both laws in the stated model for every x , not merely at plotted coordinates.

Under the alternative, $\zeta = \rho = (1+h)/2$ and $\pi = 1/2$. Since $\mu_2 = 128/3465$, their population quantities are

$$E[h(X)^2] = \mu_2 \delta^2 = \frac{2}{3465}, \quad N = -\frac{1}{6930}, \quad \Lambda = \frac{3}{640}, \quad \tau = -\frac{64}{2079}. \quad (6.2)$$

Under the null, $N = \tau = 0$ and $\Lambda = 3/640$ is unchanged. Both laws satisfy $\lambda_n \leq \Lambda \leq 2\lambda_n$.

To describe compression, define the population moment ratio

$$N_S = E[(Z - E[Z | S])(Y - E[Y | S])], \quad \Lambda_S = E[(Z - E[Z | S])(D - E[D | S])], \quad \tau_S = N_S / \Lambda_S, \quad (6.3)$$

where the ratio is used only when $\Lambda_S \neq 0$. Each bump has zero integral within its block, so under both laws the three S -projected nuisance means equal $1/2$. Consequently, $N_S = 0$, $\Lambda_S = 3/640$, and $\tau_S = 0$ under both laws. This compressed ratio is a moment pseudo-target: conditional IV validity given X does not automatically imply the same causal interpretation after conditioning only on S .

In fact, more than the compressed moments coincide. Integrating the joint law (4.1) over either summary cell eliminates its term linear in h . For every $y, d, z, s \in \{0, 1\}$,

$$P_{h=0}(Y = y, D = d, Z = z, S = s) = P_h(Y = y, D = d, Z = z, S = s) = \frac{1 + c\sigma_z\sigma_d}{16}. \quad (6.4)$$

Thus the entire distribution of (Y, D, Z, S) agrees, while the full- X causal targets differ. This is a consequence of the explicit family, not an additional general identification theorem.

The two full- X laws are distinct and fixed in this illustration. Their compressed-law equality does not show that they remain difficult to distinguish from full observations as n increases. Theorem 3.1 instead uses increasing resolution, a sample-size-dependent amplitude, and a mixture with signs shared across the entire sample. The figure explains a mechanism behind the construction; it does not replace that testing argument.

S.3.2 Explicit constants and the limits of the guarantee

Matching rates need not yield a useful numerical upper bound at a given sample size. Table 1 specializes the proved bounds to $\beta = 1/8$, $\alpha = 1/20$, and $J = J_n = \lceil n^{4/3} \rceil$. Write $b = b_{1/8, J}$, $v = v_{n, J}$, $e = e_{n, J, \alpha}$, and $R = b + e$, retaining the dyadic rounding in (5.10). Before the rate-level relaxation of the lower bound, (4.19) gives

$$L_{\text{bd}} = \frac{3\mu_2(1 - 2\alpha)^2}{16} \min \left\{ 1, \frac{J^{-1/4}}{6\lambda_n} \right\}. \quad (6.5)$$

Every full-model-honest procedure has expected diameter at least L_{bd} at P_n^* . For the particular construction $C_n^{1/8}$ in Supplement Section S.2, (5.15) gives the whole-model upper bound

$$U_{\text{bd}} = \min \left\{ 2, \frac{8R}{\lambda_n} + \frac{32v}{\lambda_n^2} \right\}. \quad (6.6)$$

These are different bounds with different quantifiers, not two measurements of the same procedure's mean diameter.

The exact scalar data (file `code/v8/out/r3_constant_diagnostics.csv`) also record

Compression can hide the target shift

Same valid binary-IV model: $\beta = 1/8$, $c = 3/160$, $\lambda = 1/320$. No sampled data.

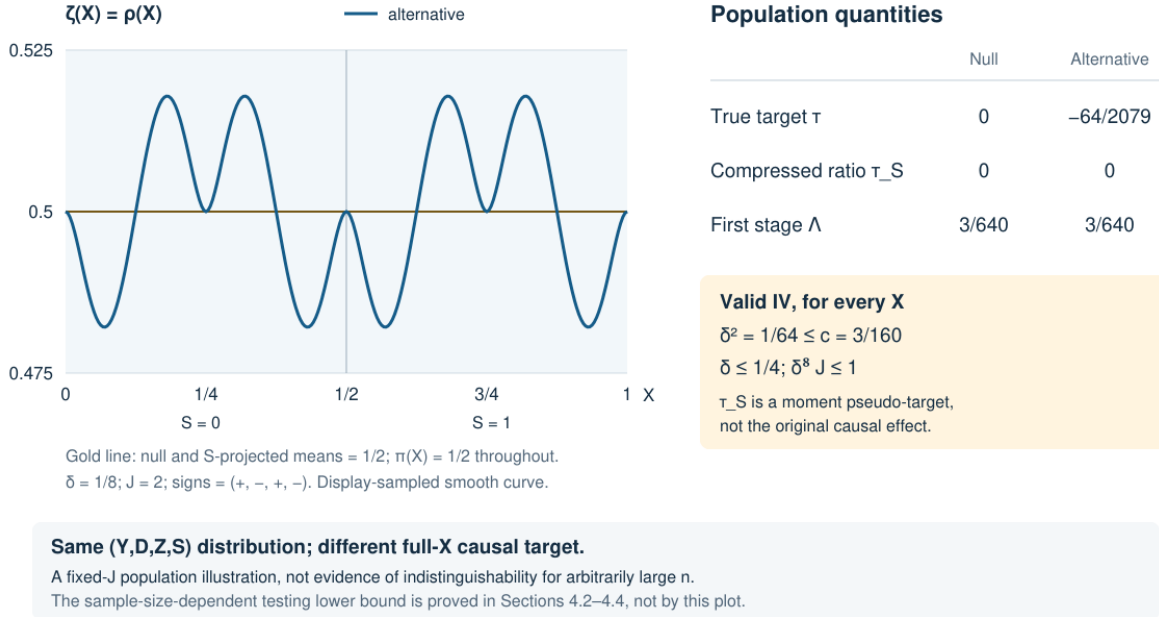


Figure 1: Compression can hide a target shift. The blue curve displays $\zeta(X) = \rho(X)$ under the alternative; the gold line is the common value of the null and S -projected means. The treatment mean is $\pi(X) = 1/2$ throughout. The two laws have the same compressed observation distribution (6.4), the same first stage $3/640$, and targets 0 and $-64/2079$. The displayed curve uses 257 coordinate points, not sampled observations; validity at every x follows analytically. This fixed- J population example is not a sample-size-dependent hard sequence. The full-observation testing lower bound is proved separately in Supplement Sections S.1.2–S.1.4. A vector version is supplied separately.

Table S.1: Deterministic evaluations of the finite-sample diameter bounds. The radius and lower bounds are exact fractions. The sole noninteger upper bound is rounded to six decimal places. No data set of the displayed size was generated.

n	λ_n	R	L_{bd} : same-null lower bound	U_{bd} : whole-model upper bound
10^3	$1/32$	$9/4$	$144/48125$	2
10^3	$1/32000$	$9/4$	$54/9625$	2
10^6	$1/32$	$29/100$	$72/240625$	2
10^6	$1/32000$	$29/100$	$54/9625$	2
10^{12}	$1/32$	$753/320000$	$18/6015625$	0.603711
10^{12}	$1/32000$	$753/320000$	$144/48125$	2

J, b, e, v and the unrounded upper bound. The stored evaluations use $\lceil n^{1/3} \rceil^4$ for resolution. All displayed sample sizes are perfect cubes, so that expression equals $\lceil n^{4/3} \rceil$ on every row. The table therefore evaluates the present construction exactly at these values; the two integer-resolution rules are not asserted to agree for general n .

For example, at $n = 10^6$, $J = 10^8$, $b = 1/25$, and $e = 1/4$, giving $R = 29/100$. Even at $\lambda_n = 1/32$, the upper bound is the trivial value two. At $n = 10^{12}$ and the same identification scale, it falls to approximately 0.603711, but the gap from the lower bound remains large. These enormous sample sizes are arguments of scalar formulas, not simulated sample counts.

An upper bound of two does not imply that every realized set equals Θ , or that its actual mean diameter equals two. Likewise, the nontrivial upper bound is not an observed or exactly attained mean. The table establishes neither practical superiority nor impossibility of improving constants. It shows that the available sufficient bounds do not by themselves justify a claim of short, practically useful confidence sets. The result is rate matching, not a sharp-constant comparison.

No Monte Carlo study accompanies these exhibits. The regime boundaries follow from the risk bounds, the compression example is a population calculation, and the table evaluates explicit inequalities. There is no claim of empirical coverage or estimator dominance requiring a simulated comparison here. A simulation could address a separately specified performance question, but would not by itself establish uniform honesty or strengthen the theoretical scope of these results.

S.4 Latent reconstruction of the perturbation family

Fix x , suppress its argument, and suppose $|h| \leq 1/4$ and $h^2 \leq c$. For the positive-direction perturbation in (4.1), the type probabilities are

$$p_{\text{AT}} = \frac{1 - c - h}{2(1 - h)}, \quad p_{\text{NT}} = \frac{1 - c + h}{2(1 + h)}, \quad p_{\text{CP}} = \frac{c}{1 - h^2}. \quad (\text{A.1})$$

Their first two numerators are positive, and $p_{\text{AT}} + p_{\text{NT}} = 1 - c/(1 - h^2)$, so the three probabilities are nonnegative and sum to one. The required outcome margins are

$$\begin{aligned}
q_{\text{AT},1} &= \frac{1-c}{2(1-c-h)}, & q_{\text{NT},0} &= \frac{1-c+2h}{2(1-c+h)}, \\
q_{\text{CP},1} &= \frac{1-h^2/c}{2}, & q_{\text{CP},0} &= \frac{1+h^2/c}{2}.
\end{aligned} \tag{A.2}$$

All denominators are positive. The bounds $1-c \pm 2h \geq 5/16$ imply that the first two ratios and their complements are nonnegative. The complier ratios belong to $[0, 1]$ by $h^2 \leq c$. Set $q_{\text{AT},0} = q_{\text{NT},1} = 1/2$ and choose a product coupling of the two potential outcomes conditional on type and X . This constructs one admissible latent law, not an additional independence restriction on the model.

Write $b_{T,d}(1) = q_{T,d}$ and $b_{T,d}(0) = 1 - q_{T,d}$. Generating the instrument independently of these latent variables given X , with $\zeta = (1+h)/2$, yields the following four identities for each $y \in \{0, 1\}$:

$$\begin{aligned}
r_{1y|0,h} &= p_{\text{AT}}b_{\text{AT},1}(y) = \frac{1-c+h(\sigma_y-1)}{4(1-h)}, \\
r_{0y|1,h} &= p_{\text{NT}}b_{\text{NT},0}(y) = \frac{1-c+h(1+\sigma_y)}{4(1+h)}, \\
r_{1y|1,h} &= p_{\text{AT}}b_{\text{AT},1}(y) + p_{\text{CP}}b_{\text{CP},1}(y) = \frac{1+c+h(1+\sigma_y)}{4(1+h)}, \\
r_{0y|0,h} &= p_{\text{NT}}b_{\text{NT},0}(y) + p_{\text{CP}}b_{\text{CP},0}(y) = \frac{1+c+h(\sigma_y-1)}{4(1-h)}.
\end{aligned} \tag{A.3}$$

For example, the complier contribution in the third line is $(c - \sigma_y h^2)/\{2(1-h^2)\}$, exactly the corresponding difference in (4.4). Multiplying each conditional cell by $(1 + \sigma_z h)/2$ recovers every joint cell in (4.1). Moreover, $\zeta(1-\zeta)p_{\text{CP}} = c/4$ and $q_{\text{CP},1} - q_{\text{CP},0} = -h^2/c$, which also recover (4.2).

Where the boundary $h^2 = c$ is compatible with $|h| \leq 1/4$, the complier margins become $q_{\text{CP},1} = 0$ and $q_{\text{CP},0} = 1$. The observed cells remain interior because of the other types' contributions. No strict interiority of latent outcome margins was assumed. Here $c > 0$, so $p_{\text{CP}} > 0$. At points with $p_{\text{CP}} = 0$ in the general model of Section 2, each nonnegative complier cell difference is zero; assigning arbitrary outcome margins to that zero-probability type leaves the observed law unchanged. This does not extend the identification stratum to $c = 0$.

S.5 Bump integrals and block boundaries

On $[0, 1]$, the polynomial in (4.5) expands as

$$\varphi(u) = 32u^5 - 80u^4 + 64u^3 - 16u^2, \quad \int_0^1 \varphi(u) du = \frac{32}{6} - \frac{80}{5} + \frac{64}{4} - \frac{16}{3} = 0. \quad (\text{A.4})$$

For nonnegative integers p, q , repeated integration by parts gives $\int_0^1 u^p (1-u)^q du = p!q!/(p+q+1)!$. Thus

$$\begin{aligned} \int_0^1 \varphi(u)^2 du &= 256 \int_0^1 u^4 (1-u)^4 (4u^2 - 4u + 1) du \\ &= 256 \left(\frac{4}{2310} - \frac{4}{1260} + \frac{1}{630} \right) = \frac{128}{3465} = \mu_2. \end{aligned} \quad (\text{A.5})$$

The factorization gives $|\varphi| \leq 1$, and its value and first derivative vanish at both endpoints. With $t = u(1-u)$, its derivative is $32t(5t-1)$: the minimum on $[0, 1/4]$ is $-8/5$ at $t = 1/10$, and the maximum is 2 at $t = 1/4$. The zero extension is therefore C^1 and globally Lipschitz with constant at most two.

For the common $m = 2J$ partition and $\omega = \delta^2$, change of variable in each block gives

$$\int_{j/m}^{(j+1)/m} h_v(x) dx = 0, \quad \int_{j/m}^{(j+1)/m} h_v(x) h_{v'}(x) dx = \frac{\mu_2 \omega}{m} v_j v'_j. \quad (\text{A.6})$$

Summing proves both the common target calculation and the inner product used in the mixture bound. In cell coordinates, the derivative within any block is bounded by $2\delta J$. The derivatives and values agree at the joins; partitioning an interval at its finitely many joins and summing the derivative bound proves the same Lipschitz bound across blocks. Finally, $\min\{1, z\} \leq z^\beta$ for $z \geq 0$ and $0 < \beta < 1$. This justifies (4.8) for every real β in the stated range. No latent-margin smoothness condition is used.

S.6 The eight-cell likelihood calculation

Let $F_c = (\sigma_Z + \sigma_Y)/(1 + c\sigma_Z\sigma_D)$. Conditional on any x , the null cell probability is $(1 + c\sigma_Z\sigma_D)/8$.

Cancellation of one denominator gives

$$\begin{aligned} E_{P^*}[F_c | X = x] &= \frac{1}{8} \sum_{\sigma_z, \sigma_d, \sigma_y} (\sigma_z + \sigma_y) = 0, \\ E_{P^*}[F_c^2 | X = x] &= \sum_{\sigma_z, \sigma_d, \sigma_y} \frac{(\sigma_z + \sigma_y)^2}{8(1 + c\sigma_z\sigma_d)} \\ &= \frac{1}{2} \sum_{\sigma_z, \sigma_d} \frac{1}{1 + c\sigma_z\sigma_d} = \frac{1}{1+c} + \frac{1}{1-c} = \frac{2}{1-c^2}. \end{aligned} \tag{A.7}$$

In the second sum only $\sigma_y = \sigma_z$ contributes; the two possibilities for each value of $\sigma_z\sigma_d$ explain the coefficients. All denominators are positive. Expanding the two likelihood ratios and applying (A.6) now gives

$$E_{P^*}[\ell_v \ell_{v'}] = 1 + \frac{2}{1-c^2} \int_0^1 h_v h_{v'} = 1 + \frac{\xi_c \omega}{m} \sum_j v_j v'_j. \tag{A.8}$$

This is the full-observation inner product, not a marginal calculation after dropping D or X . The finite-mixture expansion, independence within each sample component, and the fair-sign product identity then give (4.14). The resulting sample law remains the mixture of products (4.10), not the product of the one-observation mixture.

S.7 An optional binary-regression source

This subsection completes the alternative reduction discussed after Lemma 4.2. It is not needed for the direct proof. Put $g = h/(1-c)$, so $|g| \leq 4/13 < 1/2$, and define a source law B_g for (X, B) , where $B \in \{-1, 1\}$, by

$$B_g(B = b | X = x) = \frac{1 + 2bg(x)}{2}, \quad X \sim U[0, 1]. \tag{A.9}$$

With probability $1/2$, assign $A = B$ and draw an independent fair E ; otherwise assign $E = B$

and draw an independent fair A . This transformation, denoted L , does not depend on g . Its output law satisfies $R_g(A = a, E = e \mid X = x) = (1 + 2ag)/8 + (1 + 2eg)/8 = \{1 + g(a + e)\}/4$. Here a, e are input sign values, not the resolution exponent or confidence radius.

Next apply K_c from Supplement Section S.1.3. With probability $1 - c$ keep A, E and append a fair treatment sign; with probability c discard both input signs, draw fresh fair instrument and outcome signs, and set the treatment sign equal to the instrument sign. In both branches keep X . At each output cell,

$$(R_g K_c)(\sigma_z, \sigma_d, \sigma_y \mid x) = (1 - c) \frac{1 + g(\sigma_z + \sigma_y)}{8} + c \frac{1 + \sigma_z \sigma_d}{8} = p_h(z, d, y \mid x). \quad (\text{A.10})$$

Each branch is a probability distribution and $0 < c < 1$, so this is a Markov kernel. It depends on the fixed subexperiment scale c , but not on h . For its source-distance calculation, the likelihood ratio relative to B_0 is $1 + 2Bg(X)$. The fair null sign has mean zero and square one, hence

$$E_{B_0}[\ell_g \ell_{g'}] = 1 + 4 \int_0^1 g g', \quad \tilde{\xi}_c = \frac{4\mu_2}{(1 - c)^2} \leq \frac{131072}{585585} < 1. \quad (\text{A.11})$$

The coefficient here belongs to the binary-regression source B_g , not to the intermediate two-sign law R_g . With $g_v = h_v/(1 - c)$, the calculation in (4.14) applies with ξ_c replaced by $\tilde{\xi}_c$. In particular,

$$\begin{aligned} 1 + \chi^2 \left(2^{-m} \sum_v B_{g_v}^{\otimes n}, B_0^{\otimes n} \right) &\leq \exp \left\{ \frac{n^2 \tilde{\xi}_c^2 \omega^2}{4J} \right\}, \\ \text{TV}(Q_n, (P^*)^{\otimes n}) &\leq \text{TV} \left(2^{-m} \sum_v R_{g_v}^{\otimes n}, R_0^{\otimes n} \right) \\ &\leq \text{TV} \left(2^{-m} \sum_v B_{g_v}^{\otimes n}, B_0^{\otimes n} \right) \leq \gamma_\alpha. \end{aligned} \quad (\text{A.12})$$

For the last inequality, $\omega \leq \gamma_\alpha J^{-2\beta}$ and $J^{4\beta+1} \geq n^2$ bound the exponent by $\gamma_\alpha^2/4$; (4.15) then applies. The intermediate inequalities follow by applying the same kernels independently to the observations, while retaining one shared sign vector for the sample. The generic contraction ar-

gument is given next. This one-way reduction is neither an equivalence of experiments nor an algorithm for increasing information by generating additional instruments.

S.8 Auxiliary randomness and transfer of coverage

For probability laws Q, P and any measurable function $0 \leq f \leq 1$, the layer-cake identity and the half- L^1 convention imply $|\int f d(Q - P)| \leq \text{TV}(Q, P)$. For a Markov kernel K , apply this with $f(w) = K(w, A)$ to obtain $\text{TV}(QK, PK) \leq \text{TV}(Q, P)$.

In particular, for a common independent randomization law ν and any measurable event A in the product experiment, write $A_w = \{u : (w, u) \in A\}$. Then

$$\begin{aligned} |(Q \otimes \nu)(A) - (P \otimes \nu)(A)| &= \left| \int \nu(A_w) d(Q - P)(w) \right| \leq \text{TV}(Q, P), \\ \text{TV}(Q \otimes \nu, P \otimes \nu) &= \text{TV}(Q, P). \end{aligned} \tag{A.13}$$

The reverse inequality for the equality follows by using events of the form $A_0 \times \mathcal{U}$, where $\mathcal{U}$ is the auxiliary sample space. Coverage events are measurable by the admissibility requirement in Section 2.4; no additional regularity of an arbitrary confidence-set procedure is presumed.

For completeness, suppose a sample mixture $Q = \sum_j w_j P_j^{\otimes n}$ has component laws all with target t_1 , while a reference law P_0 has target t_0 . If coverage is at least $1 - \alpha$ at every component and the reference, averaging coverage and using (A.13) gives

$$\mathbb{E}_{P_0, n}[\text{diam}(C_n)] \geq |t_1 - t_0| [1 - 2\alpha - \text{TV}(Q, P_0^{\otimes n})]_+, \quad [a]_+ = \max\{a, 0\}. \tag{A.14}$$

Indeed, the reference probabilities of including t_1 and t_0 are at least $1 - \alpha - \text{TV}$ and $1 - \alpha$, respectively; their intersection has the stated lower bound. This proof includes disconnected sets and any auxiliary randomization. An empty set contributes zero diameter but never belongs to the joint-inclusion event. Applying (A.14) to $Q_n, t_1 = -\Delta, t_0 = 0$ gives Supplement Section S.1.4;

applying it to the single oracle alternative gives Supplement Section S.1.5. No target is assigned to the sample mixture itself.

S.9 The oracle cell table and distance

For the latent family (4.22), direct addition of always-taker, never-taker, and complier contributions gives the observed conditional cell probabilities reported in Table S.2. The probabilities do not depend on x .

Table S.2: Observed conditional cell probabilities for the oracle family in (4.22). Each entry is $P_\theta(D = d, Y = y \mid Z = z, X = x)$; the probabilities do not depend on x .

(D, Y)	$Z = 0$	$Z = 1$
$(0, 0)$	$(1 + c)/4 + c\theta/2$	$(1 - c)/4$
$(0, 1)$	$(1 + c)/4 - c\theta/2$	$(1 - c)/4$
$(1, 0)$	$(1 - c)/4$	$(1 + c)/4 - c\theta/2$
$(1, 1)$	$(1 - c)/4$	$(1 + c)/4 + c\theta/2$

Each column sums to one. For $|\theta| \leq 1/2$, the varying cells are at least $1/4$, and the other cells are at least $13/64$. The two treated-cell differences are $c(1 + \sigma_y\theta)/2$, and the two untreated-cell differences in the direction of (2.2) are $c(1 - \sigma_y\theta)/2$, all nonnegative. With $P(Z = 1 \mid X) = 1/2$,

$$E[D \mid Z = 0, X] = (1 - c)/2, \quad E[D \mid Z = 1, X] = (1 + c)/2, \quad (\text{A.15})$$

$$E[Y \mid Z = 0, X] = 1/2 - c\theta/2, \quad E[Y \mid Z = 1, X] = 1/2 + c\theta/2.$$

It follows that $\zeta = \pi = \rho = 1/2$, $\Lambda = c/4$, and $N = c\theta/4$. At $\theta = 0$, multiplying the table by $1/2$ gives exactly the joint law (3.1). For general θ , the four varying joint cells change by $\pm c\theta/4$ from the common null mass $(1 + c)/8$. Thus

$$\chi^2(P_\theta^{\text{or}}, P^*) = 4 \frac{(c\theta/4)^2}{(1 + c)/8} = \frac{2c^2\theta^2}{1 + c}. \quad (\text{A.16})$$

Independence makes the second moment of the sample likelihood the n th power of its one-observation second moment, proving the second identity in (4.23). The TV choice and two-point

transfer are completed in (4.24)–(4.25). The upper bound does not restrict the whole oracle class to this table: Supplement Section S.2.5 uses only S -measurability of the three nuisance means.

S.10 Projection normalization and orthogonality

The bin indicators $\sqrt{2J}\mathbf{1}_{I_{j,k}}$ are orthonormal in $L^2(U[0,1])$, since distinct bins are disjoint and each has probability $1/(2J)$. This verifies that (5.1) is the population projection kernel. For the symmetric kernel in (5.7), each diagonal product $Z_i V_i$ appears in $n - 1$ unordered pairs. Therefore

$$\begin{aligned} \binom{n}{2}^{-1} \sum_{i < j} \frac{Z_i V_i + Z_j V_j}{2} &= \frac{1}{n} \sum_i Z_i V_i, \\ \binom{n}{2}^{-1} \sum_{i < j} \frac{K_J(X_i, X_j)}{2} (Z_i V_j + Z_j V_i) &= \frac{1}{n(n-1)} \sum_{i \neq j} K_J(X_i, X_j) Z_i V_j. \end{aligned} \tag{A.17}$$

Thus the pair average is precisely (5.2), with no diagonal term left in its second component.

Use the notation $G_1, G_2, \vartheta_{V,J}$ from Supplement Section S.2.2. Degeneracy gives, for three distinct indices,

$$\begin{aligned} E_P[G_2(W_1, W_2)G_2(W_1, W_3)] &= E_P[E_P(G_2(W_1, W_2) \mid W_1)E_P(G_2(W_1, W_3) \mid W_1)] = 0, \\ E_P[G_1(W_1)G_2(W_1, W_2)] &= 0, \\ E_P H_{V,J}^2 &= \vartheta_{V,J}^2 + 2E_P G_1^2 + E_P G_2^2. \end{aligned} \tag{A.18}$$

The conditional factorization in the first line uses independence of W_2, W_3 given W_1 . Disjoint-index terms vanish by independence and zero means. Consequently the degenerate pair average has variance $E_P G_2^2 / \binom{n}{2}$, and it is orthogonal to the first-order average. This yields exactly (5.8), also for $n = 2$, when there are no pairs sharing a single index to count.

Finally, for independent X, X' ,

$$E_P K_J(X, X')^2 = \sum_{k=0}^{2J-1} (2J)^2 P(X \in I_{J,k}) P(X' \in I_{J,k}) = 2J. \tag{A.19}$$

The last identity and (A.18) justify retaining the degenerate term in (5.6). The argument does not assume $J \leq n$, a regular empirical design matrix, or any smoothness of the conditional joint law beyond that specified in Section 2. The projection-bias calculation itself is given in (5.3)–(5.5).

S.11 Exact inversion, empty sets, and measurability

Fix any realized real estimates $\widehat{N}, \widehat{\Lambda}$ and radius $R > 0$. This argument applies to both confidence sets of Supplement Section S.2. For $\eta = 1, -1$, let the corresponding initial intervals be $[L_1^0, U_1^0] = [0, 1]$ and $[L_{-1}^0, U_{-1}^0] = [-1, 0]$. Define the two inequality coefficients by

$$a_{\eta,1} = -\widehat{\Lambda} - R\eta, \quad a_{\eta,2} = \widehat{\Lambda} - R\eta, \quad b_1 = R - \widehat{N}, \quad b_2 = R + \widehat{N}. \quad (\text{A.20})$$

The two inequalities are $a_{\eta,j}t \leq b_j$, $j = 1, 2$. A positive coefficient gives an upper bound on t , a negative coefficient a lower bound. Define

$$\begin{aligned} t_\eta &= \max \left(\{L_\eta^0\} \cup \{b_j/a_{\eta,j} : a_{\eta,j} < 0, j = 1, 2\} \right), \\ \bar{t}_\eta &= \min \left(\{U_\eta^0\} \cup \{b_j/a_{\eta,j} : a_{\eta,j} > 0, j = 1, 2\} \right). \end{aligned} \quad (\text{A.21})$$

Both extrema are over nonempty finite sets. A zero coefficient with nonnegative right-hand side imposes no restriction; with negative right-hand side it makes that half infeasible. More precisely, put

$$F_\eta = \{t_\eta \leq \bar{t}_\eta\} \cap \bigcap_{j=1}^2 \{a_{\eta,j} \neq 0 \text{ or } b_j \geq 0\}. \quad (\text{A.22})$$

On F_η retain $[t_\eta, \bar{t}_\eta]$; otherwise retain the empty set. The union of the two retained intervals is exactly the inverted set, including at $t = 0$. Divisions in (A.21) occur only on the indicated nonzero-coefficient branches. They are not denominator clipping or division by $\widehat{\Lambda}$ to form a Wald ratio.

All sign and feasibility branches are measurable, and their endpoint functions are measurable

on each branch. The resulting random closed set is measurable: for each rational open interval (p, q) , the event that a retained interval intersects it is $F_\eta \cap \{t_\eta < q, \bar{t}_\eta > p\}$. Such intervals form a countable base. Fixed-target inclusion is equally a finite union of measurable endpoint comparisons.

Let $\mathcal{E} = \{\eta : F_\eta \text{ holds}\}$. Its diameter is

$$\text{diam}(C) = \begin{cases} 0, & \mathcal{E} = \emptyset, \\ \max_{\eta \in \mathcal{E}} \bar{t}_\eta - \min_{\eta \in \mathcal{E}} t_\eta, & \mathcal{E} \neq \emptyset. \end{cases} \quad (\text{A.23})$$

This is measurable and in $[0, 2]$. Equality of endpoints gives a singleton; two separated intervals are not replaced by the sum of their lengths. As a purely algebraic example, $(\hat{N}, \hat{\Lambda}, R) = (2, 0, 1)$ gives $C = \{-1, 1\}$, whose diameter is two. This is an illustration of the inversion map, not a claim about the frequency or attainability of this triple under the chosen procedure. The finite-sample coverage and risk bounds do not condition on nonempty outcomes.

S.12 Resolution, rounding, and the constants

For $J = \lceil n^{a_\beta} \rceil$, $n \geq 2$, and $0 < \beta < 1/4$,

$$n^{a_\beta} \leq J \leq 2n^{a_\beta}, \quad J^{-2\beta} \geq \frac{1}{2}n^{-r_\beta}, \quad \frac{n^2}{J^{4\beta+1}} \leq 1, \quad a_\beta - 2 = -2r_\beta. \quad (\text{A.24})$$

The lower bound uses $2^{-2\beta} \geq 1/2$. For $z \geq 0$, $\min\{1, z/12\} \geq \min\{1, z\}/12$, so the multiplier in (4.19) gives

$$\frac{3\mu_2(1-2\alpha)^2}{16} \cdot \frac{1}{12} = \frac{\mu_2(1-2\alpha)^2}{64} = \frac{2(1-2\alpha)^2}{3465}. \quad (\text{A.25})$$

This verifies the lower constant for the actual rounded resolution rather than a continuous optimization.

For any $x > 0$, put $k = \lceil \log_2 x \rceil$. Then $k-1 < \log_2 x \leq k$, and hence

$$x \leq 2^k < 2x. \quad (\text{A.26})$$

At an exact power of two the first inequality is equality; the second remains strict. Negative k gives a positive dyadic fraction. Taking $x = \sqrt{2v_{n,J}/\alpha}$ proves the complete rounding claim in (5.10) without an additive numerical error floor.

For the large-model upper bound, $n-1 \geq n/2$ and $r_\beta < 1/2$ yield $v_{n,J} \leq n^{-1} + 4n^{-2} + 8J/n^2 \leq (1+4+16)n^{-2r_\beta}$. The bias and rounded stochastic allowances therefore give

$$b_{\beta,J} \leq 4n^{-r_\beta}, \quad e_{n,J,\alpha} < 2\sqrt{42/\alpha}n^{-r_\beta}, \quad 16R_{\beta,n,J} \leq 16(4+2\sqrt{42/\alpha})n^{-r_\beta}. \quad (\text{A.27})$$

Combining this with (5.15), using $\min\{1, Az\} \leq A \min\{1, z\}$ for $A \geq 1$, verifies the upper constant in (3.4). This combination relies on the bad-denominator calculation in Supplement Section S.2.4, not just on the coverage event.

For the oracle, the exact zero bias at $J = 1$ gives $v_{n,1} \leq 7/n$. Thus one pair of admissible constants in (3.9) is

$$\underline{k}_{0,\alpha} = \frac{1-2\alpha}{2} \min\{1/2, b_\alpha\}, \quad \bar{k}_{0,\alpha} = 32\sqrt{14/\alpha}, \quad (\text{A.28})$$

where b_α is defined in (4.24). To check the lower constant, $2nc^2\theta_n^2/(1+c) \leq 72b_\alpha^2$ gives sample TV at most $\frac{1}{2}\sqrt{\exp(72b_\alpha^2)-1} = \varepsilon_\alpha$; (A.14) and the minimum comparison stated after (4.25) apply. The upper constant follows from (A.26) and the zero-bias version of (5.15). Both are positive and finite, with the displayed lower constant smaller than the upper one.

These calculations complete the supplementary details for Theorem 3.1 and Proposition 3.2 under the stated finite-sample contract. The contraction statements in Section 3.4 are consequences of their risk bounds and the bounded diameter. The formulas define mathematical procedures for known real β ; no exact finite-precision implementation for arbitrary real inputs, adaptation to unknown β , or uniform asymptotic-honesty extension is asserted.

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